Thyroid & parathyroid

Lobes, capsules, relations, arteries, veins, the recurrent and external laryngeal nerves, and the parathyroid glands.

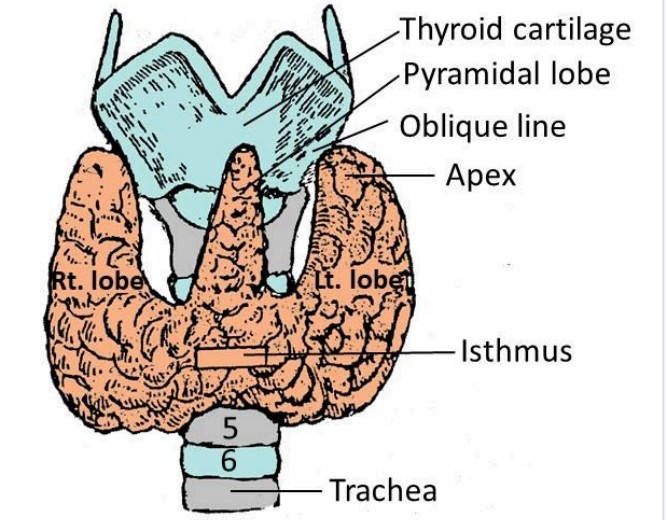
1. Gross anatomy
2. Relations of the lateral lobe
3. Arterial supply
4. Venous drainage
5. Parathyroid glands
6. Development
7. In theatre
8. On imaging

01 Gross anatomy

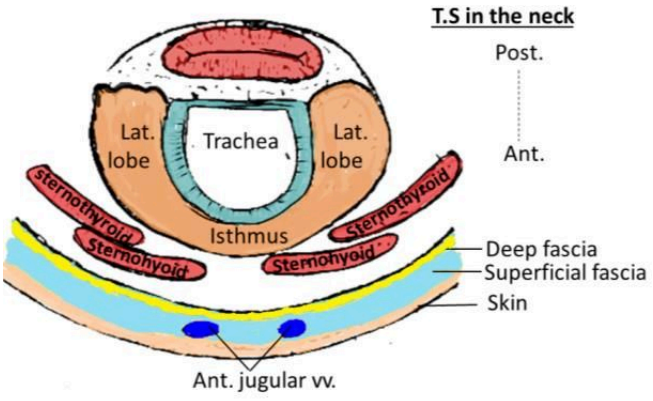
- Two **lateral lobes** joined by the **isthmus** over **tracheal rings 2–4**. Lobes reach from the oblique line of the thyroid cartilage to about the 6th tracheal ring.
- **Pyramidal lobe** from the isthmus toward the hyoid in many people: a remnant of the thyroglossal duct.
- **Two capsules:** a **true capsule** (condensed stroma) and a **false capsule** from the pretracheal fascia. Vessels run between them.
- Attached to the cricoid and trachea by the **ligament of Berry**, so the gland moves with swallowing.

CLINICAL PEARL Thyroglossal duct cysts rise on swallowing and tongue protrusion; Sistrunk's operation removes the central hyoid with the tract up to the foramen caecum.

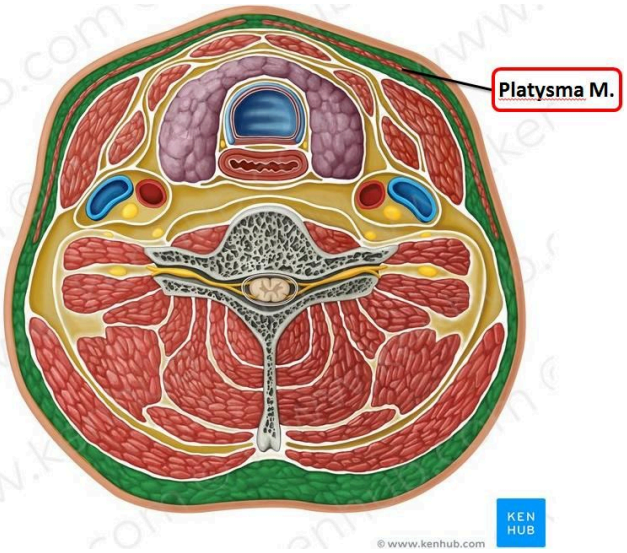
SOURCES DIFFER LOWER EXTENT OF THE LOBES: Doss says the 5th or 6th tracheal ring.



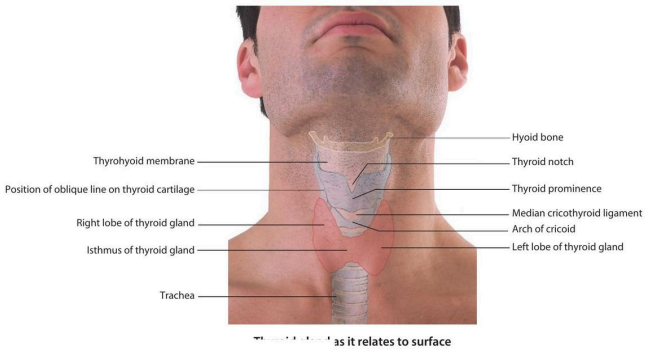
Thyroid gland, anterior view
Dr. Sameh Doss handout, p. 124



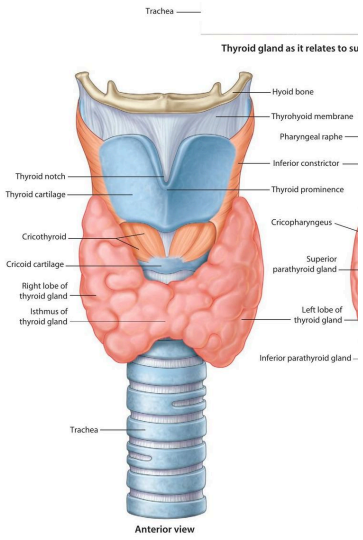
Thyroid gland in cross-section with its capsules
Dr. Sameh Doss handout, p. 125



Cross-section of the neck at the thyroid level
ENT Steps Anatomy, p. 273



Thyroid gland as it relates to the surface
Gray's Atlas of Anatomy, 2nd ed., p. 563

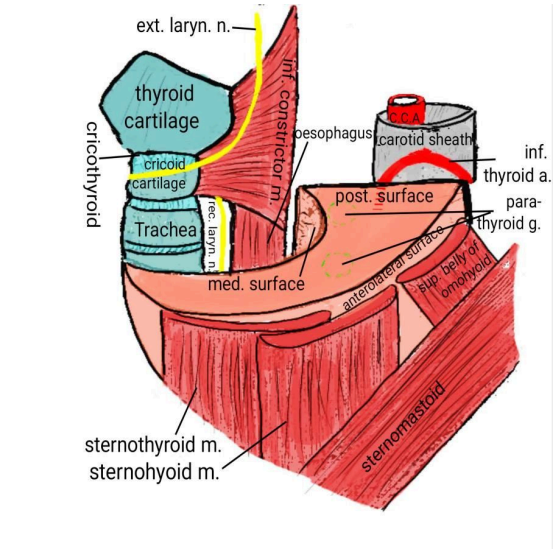


Thyroid gland (anterior view)
Gray's Atlas of Anatomy, 2nd ed., p. 563

02 Relations of the lateral lobe

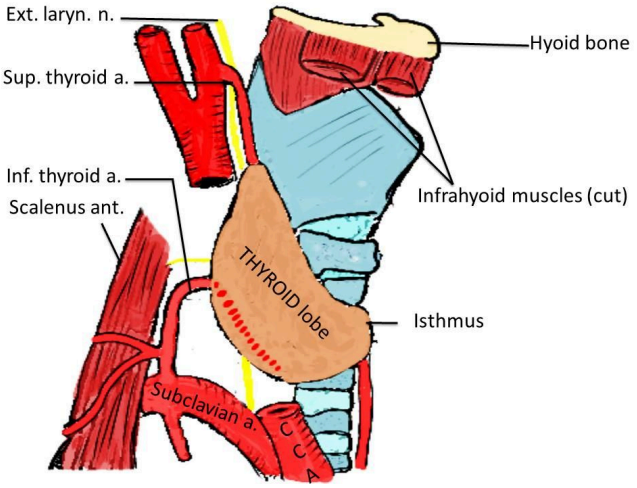
- **Anterolateral:** strap muscles (sternothyroid, sternohyoid, superior belly of omohyoid) and SCM.
- **Medial:** larynx, trachea, pharynx, oesophagus, the **external laryngeal nerve** above and the **recurrent laryngeal nerve** below in the tracheo-oesophageal groove.
- **Posterolateral:** carotid sheath.
- **Posterior border:** parathyroid glands and the anastomosis of superior and inferior thyroid arteries.
- **Tubercle of Zuckerkandl:** posterior projection of the lobe; the RLN usually runs medial (deep) to it.

> **CLINICAL PEARL** The RLN is most at risk where it passes under the ligament of Berry and near the tubercle of Zuckerkandl.



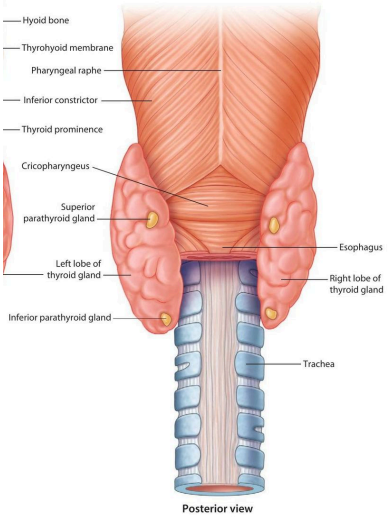
Relations of the lateral lobe

Dr. Sameh Doss handout, p. 126



Thyroid with infrahyoid muscles reflected

Dr. Sameh Doss handout, p. 127



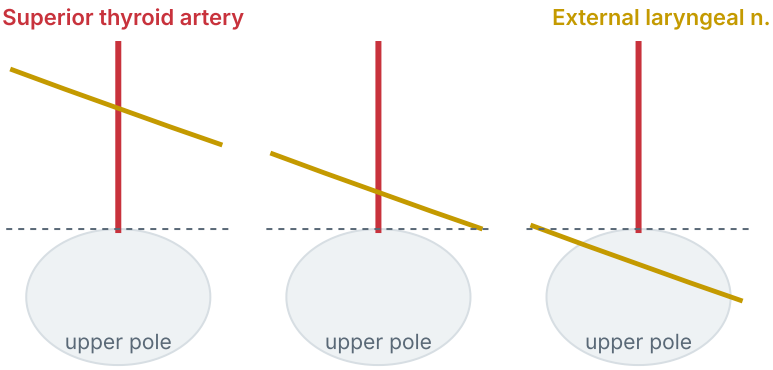
Thyroid gland (posterior view)

Gray's Atlas of Anatomy, 2nd ed., p. 563

03 Arterial supply

- **Superior thyroid artery** (first branch of ECA) to the upper pole, with the **external laryngeal nerve** nearby.
- **Inferior thyroid artery** (thyrocervical trunk) to the lower back of the lobe, **crossing the RLN** (in front, behind or between its branches).
- **Thyroid ima artery** in a small minority (about 3–10%), from the brachiocephalic trunk or arch, reaching the isthmus.
- **Cernea classification of the external laryngeal nerve:** type 1 crosses the superior thyroid artery more than 1 cm above the upper pole; type 2a less than 1 cm above; type 2b below the upper border of the pole (highest risk).

> **CLINICAL PEARL** Ligate the superior pole vessels close to the gland, branch by branch, to spare the external laryngeal nerve.

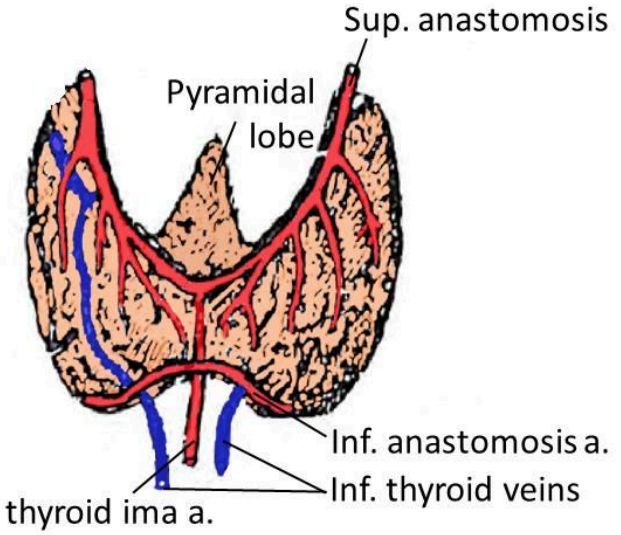


Type 1
crosses > 1 cm above

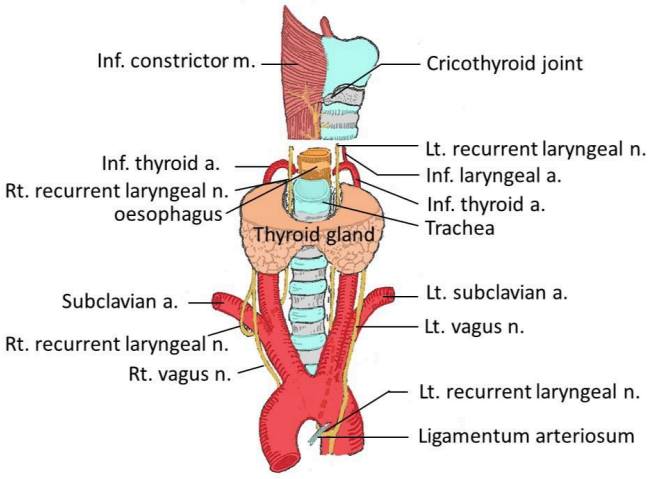
Type 2a
crosses < 1 cm above

Type 2b
crosses below pole top

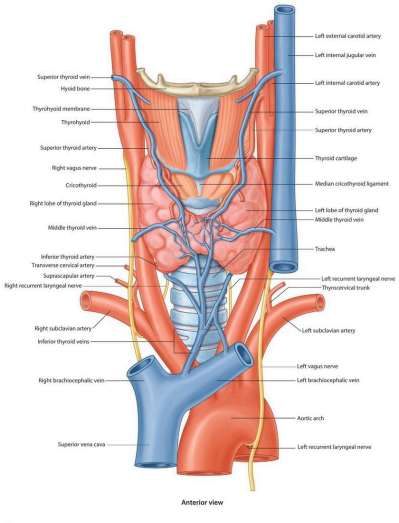
Cernea classification. The lower the nerve crosses the superior thyroid vessels, the higher the risk at upper-pole ligation; type 2b is the most dangerous.



Arterial supply of the thyroid
Dr. Sameh Doss handout, p. 125



Recurrent laryngeal nerves and inferior thyroid arteries
Dr. Sameh Doss handout, p. 154



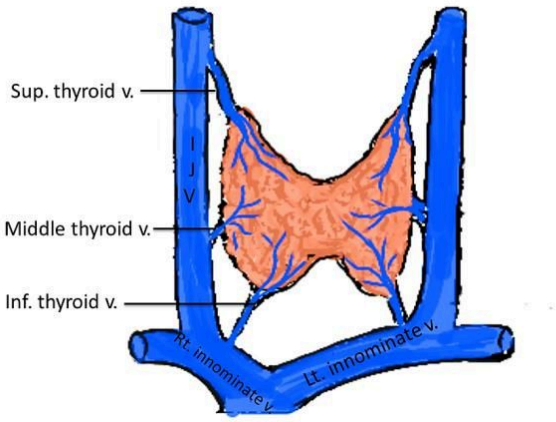
Vasculature of the thyroid gland (anterior view)
Gray's Atlas of Anatomy, 2nd ed., p. 564

04 Venous drainage

- **Superior thyroid vein** → IJV (with the artery).

- **Middle thyroid vein** → IJV (no accompanying artery; tie it first when mobilising the lobe).
- **Inferior thyroid veins** → brachiocephalic veins, forming a plexus in front of the trachea.

> **CLINICAL PEARL** The inferior thyroid veins in front of the trachea bleed at a low tracheostomy.



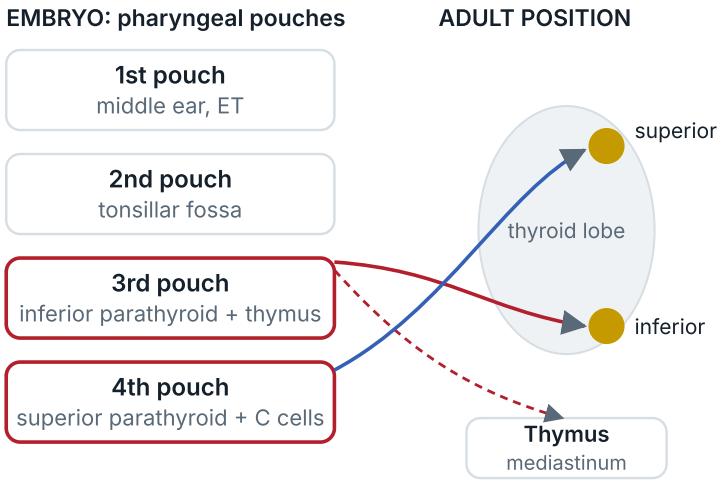
Veins of the thyroid
Dr. Sameh Doss handout, p. 128

05 Parathyroid glands

- Usually four, about 5 × 3 mm, yellow-brown.
- **Superior parathyroids** from the **4th pharyngeal pouch**: more constant, at the back of the upper lobe near the cricothyroid junction, **dorsal (posterior) to the RLN**.
- **Inferior parathyroids** from the **3rd pouch** (with the thymus): more variable (lower pole, thyrothymic ligament, thymus, even mediastinum), **ventral (anterior) to the RLN**.
- Blood supply mainly from the **inferior thyroid artery**.

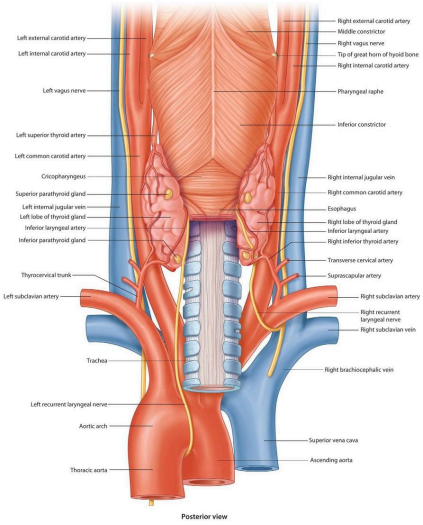
> **CLINICAL PEARL** Because the inferior glands travel with the thymus, an ectopic inferior adenoma is often in the thymus or anterior mediastinum.

> **SOURCES DIFFER SIZE:** Doss 6 × 3 mm. **SUPERIOR GLAND:** Doss puts it at the middle of the posterior surface of the lobe; surgical texts at the cricothyroid junction.



The 3rd pouch starts higher but its gland ends lower:
pulled down by the thymus, so inferior glands are the ectopic ones.

Why the inferior parathyroid comes from the 3rd pouch: it descends with the thymus past the 4th-pouch gland. Look for missing inferior glands in the thymus.



Vasculature of the thyroid gland (posterior view): parathyroids and recurrent laryngeal nerves
Gray's Atlas of Anatomy, 2nd ed., p. 565

06 Development

- Thyroid begins at the **foramen caecum** and descends through the **thyroglossal duct** in front of the hyoid to the neck.
- **C cells** (calcitonin) come from the **ultimobranchial body** of the 4th/5th pouch, so they sit mainly in the upper-middle parts of the lobes.
- Failure of descent gives a **lingual thyroid**; it may be the only thyroid tissue present.

> **CLINICAL PEARL** Check for other thyroid tissue with a scan before removing a lingual thyroid or thyroglossal cyst.

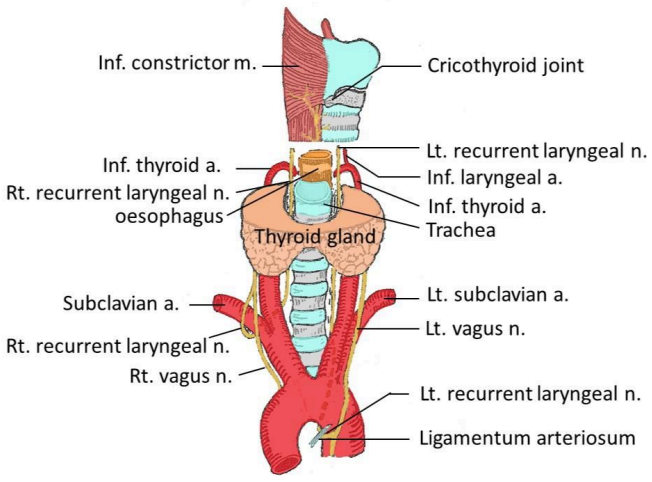
07 Surgical anatomy: In theatre

- **Order:** collar incision → subplatysmal flaps → split strap muscles in the midline → ligate the middle thyroid vein → superior pole vessels close to the gland (spare the external laryngeal nerve) → find the

RLN (inferior thyroid artery crossing, tracheo-oesophageal groove, or Zuckerkandl's tubercle) → preserve parathyroids on their blood supply → divide Berry's ligament carefully.

- **RLN identification approaches:** lateral (at the inferior thyroid artery), inferior (in the tracheo-oesophageal groove, at Simon's triangle), superior (at its laryngeal entry).
- **Nerve monitoring** confirms function; loss of signal on the first side may justify staging the second side.

> **CLINICAL PEARL** Bilateral RLN palsy after total thyroidectomy causes stridor at extubation because both cords lie paramedian.



Recurrent laryngeal nerves and inferior thyroid arteries

Dr. Sameh Doss handout, p. 154

08 Radiological anatomy: On imaging

- **Ultrasound** grades nodules (e.g. TI-RADS): suspicious features are solid hypoechoic, taller-than-wide, irregular margins, microcalcification, extrathyroidal extension.
- **Radionuclide scan:** hot nodules are rarely malignant; useful when TSH is low.
- **Parathyroid localisation:** sestamibi scan and ultrasound; 4D-CT when they disagree.

T Tables

Superior vs inferior parathyroids

	Superior	Inferior
Origin	4th pharyngeal pouch	3rd pharyngeal pouch (with thymus)
Position	Back of upper lobe, near cricothyroid junction	Near lower pole; variable
Relation to RLN	Dorsal (deep)	Ventral (superficial)
Ectopic sites	Retro-oesophageal, tracheo-oesophageal groove	Thymus, anterior mediastinum, carotid sheath
Blood supply	Inferior thyroid artery (mostly)	Inferior thyroid artery

Pharyngeal arch derivatives

Arch	Nerve	Skeletal	Muscles
1st	V3	Malleus (head), incus (body), Meckel's cartilage, sphenomandibular ligament	Mastication, tensor tympani, tensor veli palatini, mylohyoid, anterior digastric

Arch	Nerve	Skeletal	Muscles
2nd	VII	Stapes (superstructure), styloid, stylohyoid ligament, lesser horn and upper hyoid body	Facial expression, stapedius, stylohyoid, posterior digastric
3rd	IX	Greater horn and lower hyoid body	Stylopharyngeus
4th	X (superior laryngeal)	Thyroid cartilage	Cricothyroid, pharyngeal constrictors, levator veli palatini
6th	X (recurrent laryngeal)	Cricoid, arytenoid, corniculate	Intrinsic laryngeal muscles (except cricothyroid)

Pouches: 1st → middle ear and ET; 2nd → tonsillar fossa; 3rd → inferior parathyroid and thymus; 4th → superior parathyroid and C cells.

Pharyngeal pouch derivatives

Pouch	Derivative
1st	Middle ear cavity, Eustachian tube, mastoid antrum
2nd	Tonsillar fossa (crypta magna), palatine tonsil epithelium
3rd	Inferior parathyroid (dorsal), thymus (ventral)
4th	Superior parathyroid
4th/5th (ultimobranchial body)	Parafollicular C cells

Blood supply of the thyroid

Vessel	Source / drainage	Nerve at risk
Superior thyroid artery	ECA (first branch)	External laryngeal nerve
Inferior thyroid artery	Thyrocervical trunk	Recurrent laryngeal nerve
Thyroid ima artery	Brachiocephalic trunk or arch (3–10%)	—
Superior thyroid vein	IJV	—
Middle thyroid vein	IJV (no artery)	—
Inferior thyroid veins	Brachiocephalic veins	—

M Mnemonics

Parathyroid origins

Inferior from three, superior from four

Inferior glands from the 3rd pouch travel further and lower than superior glands from the 4th

Thyroid veins

Superior and middle go to the jugular, inferior goes to the brachiocephalic

Middle thyroid vein has no artery

RLN and inferior thyroid artery

Nerve near the artery

The RLN crosses the inferior thyroid artery near the lower pole: find it there

Cernea

1 safe, 2a careful, 2b danger

Type 2b crosses below the upper pole top

Q Rapid recall

1. **Which tracheal rings lie behind the thyroid isthmus?** Rings 2–4.
2. **Which thyroid vein has no accompanying artery?** Middle thyroid vein, draining to the IJV.

3. **Origin of the superior and inferior parathyroids?** Superior from the 4th pouch, inferior from the 3rd pouch with the thymus.
4. **Relation of superior and inferior parathyroids to the RLN?** Superior dorsal to the nerve; inferior ventral to it.
5. **Cernea type at highest risk of external laryngeal nerve injury?** Type 2b: crosses below the upper border of the superior pole.
6. **What is the ligament of Berry?** Posterior suspensory ligament fixing the thyroid to the cricoid and upper trachea; the RLN passes close to it.
7. **Main blood supply of the parathyroids?** Inferior thyroid artery.
8. **Two capsules of the thyroid?** True capsule (stroma) and false capsule (pretracheal fascia).
9. **Why does the thyroid move on swallowing?** It is attached to the larynx by the pretracheal fascia and Berry's ligament.
10. **What is the tubercle of Zuckerkandl?** A posterior projection of the lateral lobe; the RLN usually lies medial to it.
11. **Frequency of the thyroid ima artery?** About 3–10%.
12. **Where do the inferior thyroid veins drain?** Brachiocephalic veins.
13. **Where do C cells come from?** Ultimobranchial body (4th/5th pouch).
14. **Three ways to find the RLN at thyroidectomy?** Lateral (at the inferior thyroid artery), inferior (tracheo-oesophageal groove), superior (laryngeal entry).
15. **Result of bilateral RLN injury?** Both cords paramedian: stridor and airway obstruction, often at extubation.

Figures from ENT Steps Anatomy (based on Bansal) and Dr. Sameh Doss's Anatomy of Head & Neck handout, for personal study.